

Alexander Radovich

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EDUCATION

Colgate University

Hamilton, NY

Bachelor of Arts, Double Major: Applied Mathematics and Computer Science

Expected: May 2027

GPA: 3.8/4.00 | Dean's Award for Academic Excellence with Distinction (All 6 Semesters)

Relevant Coursework: Real Analysis, Linear Algebra, Numerical Analysis, Introduction to Statistics, Graph Theory, Data Structures & Algorithms, Computational Mathematics

EXPERIENCE

Machine Learning Research Fellow

May 2026 – Aug. 2026

University of Maryland – GAMMA Lab, Dr. Ming Lin

College Park, MD

- Developed an inference-time contrastive sampling method for video LLMs that constructs contrastive samples from vision-encoder output tokens rather than re-encoding perturbed frames, cutting contrastive decoding overhead 68% ($1.31\times$ vs. $1.97\times$ base inference cost) against the SEASON baseline on Qwen2.5-VL-7B.
- Outperformed SEASON by 16 points on MVBench temporal reasoning (33.5 to 49.5) while remaining above the unmodified model, placing the method on the accuracy/latency frontier where the frame-level baseline is dominated; confirmed on Qwen3-VL.
- Built the evaluation pipeline in VLMEvalKit across MVBench and TempCompass on a single NVIDIA A6000, using Slurm and Docker for reproducible runs.
- Manuscript in preparation, targeting CVPR 2027.

Machine Learning Engineering Intern

Jun. 2025 – Aug. 2025

DMEA

Montrose, CO

- Trained an MLP regressor on 50+ engineered features across 100k+ row datasets to forecast revenue; validated on a held-out future period in addition to cross-validation before deploying to production for live financial planning ($R^2 = 0.98$ on held-out data).
- Handled end-to-end feature engineering including one-hot encoding, Yeo-Johnson normalization, and domain-specific transformations across features; reduced manual workflow overhead by 60% through automated preprocessing pipelines.

PROJECTS

Regime-Conditional KAN Forecasting | *Python, PyTorch, KAN, HMM, NumPy*

Aug. 2025 – Present

- Built a regime-conditional forecasting system adapting the KASPER concept with a Gaussian HMM regime detector in place of the original Gumbel-Softmax gate, fitting one Kolmogorov-Arnold Network per market state across SPY, QQQ, GLD, and TLT.
- Trained 30 KANs and 10 HMMs under a 10-fold walk-forward protocol (1,008-day rolling train window, 252-day step, 3,753 feature rows over 15 years); diagnosed look-ahead bias in the training pipeline that had inflated results 16x and corrected reported Sharpe from 6.28 to 0.38.
- Post-hoc analysis of the learned spline activations surfaced a conditional signal a single global model averages away: TLT momentum predicts next-day SPY returns more reliably than equity momentum in low-volatility bull regimes.

PicGate | *Full-Stack Web Application* | picgate.vercel.app

Jan. 2025 – Present

- Built and deployed a full-stack image sharing platform for Colgate's 102-member photography club, handling uploads, gallery management, and role-based access.

TECHNICAL SKILLS

Languages: Python, C++, C, SQL, Java, MATLAB, $\text{\LaTeX}$

ML / Deep Learning: PyTorch, HuggingFace Transformers, Vision-Language Models, Contrastive Decoding, Attention Mechanisms, KV-Cache, Mixed-Precision Inference, scikit-learn, XGBoost, hmmlearn, pykan

Evaluation & Infrastructure: VLMEvalKit, Slurm, Docker, CUDA, Git/GitHub, Linux, Kubernetes, CI/CD

Data & Tools: NumPy, Pandas, Matplotlib, PostgreSQL, FastAPI, Jupyter Notebooks